

“EMBEDDED WITHIN NETWORK DEVICES, AI AGENTS CAN DETECT ANOMALIES, PREDICT POTENTIAL FAILURES, AND IMPLEMENT CORRECTIVE ACTIONS PROACTIVELY WITHOUT HUMAN INTERVENTION. THIS AUTONOMY ALLOWS NETWORKS TO ADAPT DYNAMICALLY TO CHANGING CONDITIONS AND USER DEMANDS, LEADING TO ENHANCED RELIABILITY AND PERFORMANCE.”

— **PRAMOD GUMMARAJ, FOUNDER & CEO, APRECOMM**



broadband demand, Excitel’s subscriber base increased by 55%. Consequently, there was also a rise in support calls, but support staff could not move around safely due to the raging pandemic. Excitel managed the situation by deploying Aprecomm’s agentic AI-driven Virtual Wireless Expert (VWE), a cloud-based upgrade seamlessly deployed to the Wi-Fi access points of around 500,000 subscribers. Using advanced AI algorithms, VWE enabled real-time monitoring, remote management, and autonomous self-healing.

Excitel reported a 35% improvement in first-call resolution, a 30% reduction in support call duration, and a 62% drop in truck rolls. This enhanced customer experience and network reliability resulted in a dramatic growth in Excitel’s subscriber base to over one million.

“The idea of agents is not new; we have been working on this for a while,” said Thadeous Goodwyn, director of generative AI at Booz Allen Hamilton, in a media interview. “But the reason why it’s getting so much attention now is because large language models and generative AI accelerated some of the characteristics agentic AI needs to be successful.”

Companies—not just large enterprises, but also small and medium-sized ones—across the world have started deploying AI agents for various jobs in software

development, customer service, business analytics, human resource management, etc. Interestingly, we even see AI agents being employed in chip design, for activities like code generation, verification, and optimisation.

Deloitte’s State of Generative AI in the Enterprise, published in January 2025, names agentic AI as one of the most closely watched areas in AI development. 52% of the respondents picked agentic AI, while 45% chose multi-agent systems as the most interesting area in AI now.

So, what is this trend all about? Let us attempt to simplify and understand what is agentic AI, how it is different from generative AI, what are AI agents, what kind of tasks can they handle, how do they work, what are their components, how are companies across the world using them, future trends, and challenges, and more.

Say hello to agentic AI and AI agents

We see a lot of terms like generative AI (gen AI), agentic AI and AI agents strewn around the web, often irresponsibly, without realising that they are different in terms of their core functionality and objectives. So, before we even begin to understand the impact of agentic AI, let us first try to decipher these terms.

Generative AI. Gen AI is primarily focused on creating content such

as text, images, video, audio, or software code in response to a user’s prompt. It uses machine learning (ML), large language models (LLM), and reinforcement learning (RL) to study a vast amount of given data and identify patterns and relationships in that. It uses that learning to understand natural language prompts and generate new content in real time.

Agentic AI. An agentic AI system is focused on completing tasks autonomously to achieve a given goal, rather than just creating content or answering questions. It is equipped with advanced reasoning capabilities, and can autonomously make decisions and complete the given job, even in ad hoc situations, with limited or no supervision.

Rather than learn from a prescribed dataset, an agentic AI system learns on the go from the ecosystem it works in. It constantly gathers data, and enhances its knowledge and methods in real time. It can independently analyse a situation, decide what needs to be done, and act to achieve its goal, while also learning from that experience to fine-tune itself.

The key point to be noted is that agentic AI is not about executing single, clearly-defined tasks, but about achieving a goal, which might involve strategising and executing multiple steps or tasks. An agentic AI system can work within an existing organisational setup as part of a chain of command and interact with external tools, databases, and application programming interfaces (APIs).

AI agents. AI agents can be seen as the building blocks of an agentic AI framework. An AI agent is a simpler task-oriented system with limited scope and functionality, vis-à-vis a broader, goal-oriented agentic AI system.